LAB EXPERIMENT 5

Experiment 5: Word Stemming Using Porter Stemmer

Aim

To perform word stemming on a list of words using the Porter Stemmer algorithm provided by NLTK.

Algorithm

1. Start the program.
2. Import PorterStemmer from nltk.stem.
3. Create a PorterStemmer object.
4. Define a list of words to be stemmed.
5. Pass each word to the stem() method.
6. Display each original word and its stem.
7. Stop the program.

Code

```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()
words = ["playing", "played", "plays", "running", "happily", "studies"]

print("Original Word -> Stem")
for word in words:
    print(word, "->", stemmer.stem(word))
```

Output / Screenshot

```
from nltk.stem import PorterStemmer

ps = PorterStemmer()

words = ["playing", "played", "plays", "running", "happily", "studies"]

for word in words:
    print(word, "->", ps.stem(word))
```

```
playing -> play
played -> play
plays -> play
running -> run
happily -> happili
studies -> studi
```

Result

Thus, the Porter Stemmer algorithm was applied successfully to the given words to produce their stems.